Higher National Diploma in Information Technology

First Year, First Semester Examination - 2022

HNDIT1012 - Visual Application Programming

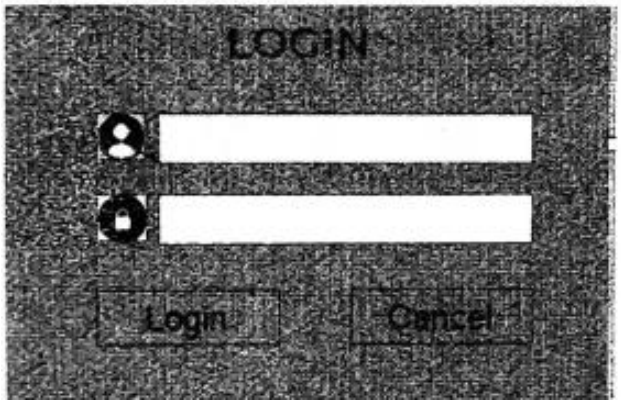
Question 01

1. Define the following terms (4 Marks)

   a. Programming Language

   b. Computer Programming

2. Following is a sample login form that is used for a simple Windows Form Application.



a. Name the components required to create the above GUI. (04 Marks)

b. Suggest suitable names for each component you have used. (04 Marks)

c. Write a C# program to implement the given functionalities when user clicks Login button.

i. Display the form named "frmTemperatureConverter" when user enters a valid user name and a password. (assume the valid username is: Admin and password is: admin@1234). (04 Marks)

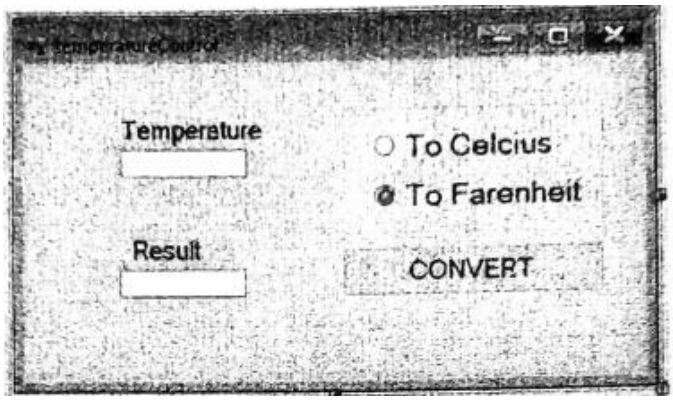
ii. When the username or password is incorrect, displays an error message stating "Invalid username or password!". (02 Marks)

d. Write the code to Exit the application when user clicks Cancel button. (02 Marks)

Question 02

1. Briefly explain the following by giving suitable examples for each.
   a.Sequential structures
   b. Selection structures (Conditional)(6 Marks)
   c.Repetition/Iterative structures (Loop)

2. Following is a Graphical User Interface (GUI) that can convert a given temperature to Celsius or fahrenheit. (08 Marks)



a. What is the purpose of using GroupBox control in the above GUI
b. Name property is an important property used to identify a GUI control/component. Give another property of each of the following controls that can be used used for the above application (04 Marks)
   i. Form
   ii. TextBox
   iii. RadioButton
   iv. Button
c. Write C# code to convert a given temperature to a required unit.
   (Hint: C(F-32)5/9; Where F-Fahrenheit and C-Celsius) (08 Marks)

Question 03

1. Briefly explain about "Events" in C# programming giving at least two examples. (04 Marks)

2. Write methods that performs the given tasks.

i. Display a message indicating which mouse button (left/right) is clicked on a form (04 Marks)

ii. Sort set of values in a given string array when user clicks a button (04 Marks)

2. Write sample programs to perform the following tasks
    i. Check two given integers, and return true if one of them is 1000 (04 Marks)
    ii. Calculate the sum of values in a given array of integers. (04 Marks)

Question 04

1. Briefly explain about Verbatim String in C# with appropriate examples. (04 Marks)
2. **Briefly explain the importance of using comments in programming** giving examples for different types of comments in C#. (04 Marks)
3. Give four (04) best practices of naming a variable. (04 Marks)
4. Give output of the following code segments (08 Marks)

i.

```
int x=0;
bool b = x>5;
if(!b)
   {
            Console.WriteLine(b);
   }
```

ii.

```
Char[] letters = { 'A', 'B', 'C'};
for(int i = 1; i < letters.Length; i++)
  {
     Console.Write(letters[i]+"\t");
  }
```

iii.

```
int i = 10;
  while (i >5)
  {
     --i;
     Console.WriteLine(i);
  }
```

iv.

```
int i = 1;
  do
  {
  Console.WriteLine(++i);
  } while (i <4);
```

Question 04

1. What is an **Array**? (04 Marks)
2. What are the two types of arrays used in C#? Explain each with a suitable example. (06 Marks)
3. Write a C# code segment to find the **sum of elements** in an integer array of size 10. (06 Marks)
4. What is a **Method**? State two (02) benefits of using methods in C# programming. (04 Marks)

Question 05

1.Briefly explain about the purpose of Language translator software, giving at least two examples. (04 Marks)

2. Give four (04) methods of a Graphics object that can be used for drawing shapes and lines. (04 Marks)

3. Briefly explain the term "DataSet" in C#. (04 Marks)

4. Write a C# program to draw two blue and red color lines in parallel with a thickness of 2. (08 Marks**)**

Question 06

1. State the purpose of following properties of FileDialog. (04 Marks)

   a. FileName

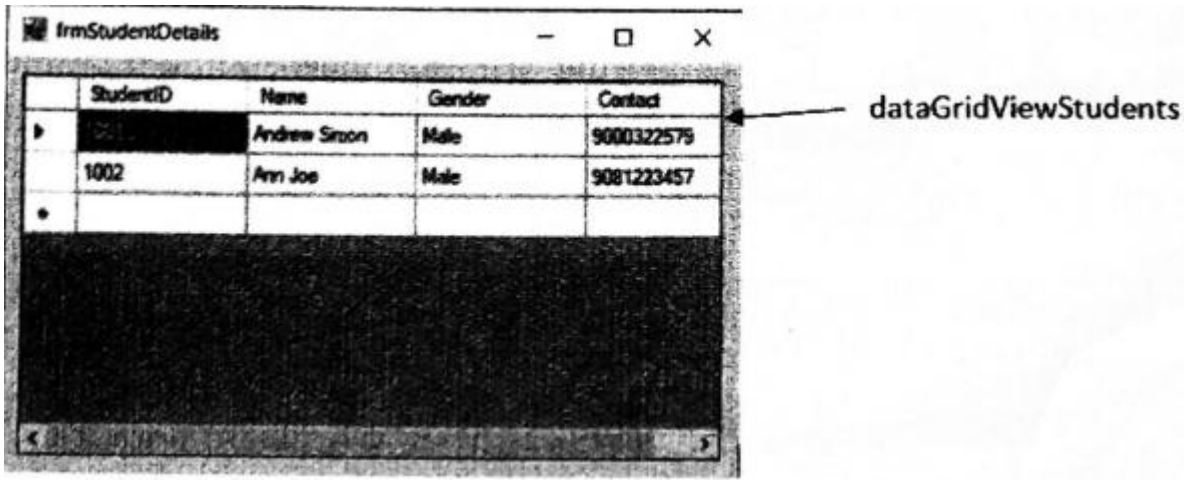
   b. Filter

   c. Initial Directory

   d. FilterIndex

2. What are the **four (04) file reading and writing methods** used in C# programming language? (04 Marks)

3. Following is a sample program outline for creating the given data table. Fill in the blanks with appropriate code segments. (12 Marks)




```
private void frmStudentDetails_Load(object sender, EventArgs e)
{
   // creating DataTable
   ......(a)...... StudentDetails = ......(b)......("StudentDetails"); // 2 marks

   //to create the column and schema
```

```
    DataColumn stdID = new DataColumn("StudentID", ......(c).....); // 1 mark
    StudentDetails.Columns.Add(stdID);
    DataColumn stdName = new DataColumn("Name", typeof(string));
    StudentDetails.Columns.Add(stdName);
    DataColumn stdGender = new DataColumn("Gender", ......(d)......); // 1 mark
    StudentDetails.Columns.Add(stdGender);
    DataColumn stdContact = new DataColumn("Contact", typeof(string));
    StudentDetails.Columns.Add(stdContact);

    //Add the Data
    ......(e)...... ;…….. 1 mark
    ......(f)...... ;…….. 1 mark
    // create dataset
.........(g)..........;…….. // 2marks
//Adding DataTables into DataSet
.........(h)..........; ……// 2marks
.........(i)..........DataSource = ......(j)..........;  …….// 2marks
}
```